Revolutionizing Online Retail: AI Virtual Try-On

Our AI-driven virtual try-on model is transforming the way people shop for clothes, glasses and jewellery. This cutting-edge technology is making online shopping easier and more enjoyable for shoppers around the world.

We provide an innovative solution for online shoppers to try-on clothes and jewellery virtually.

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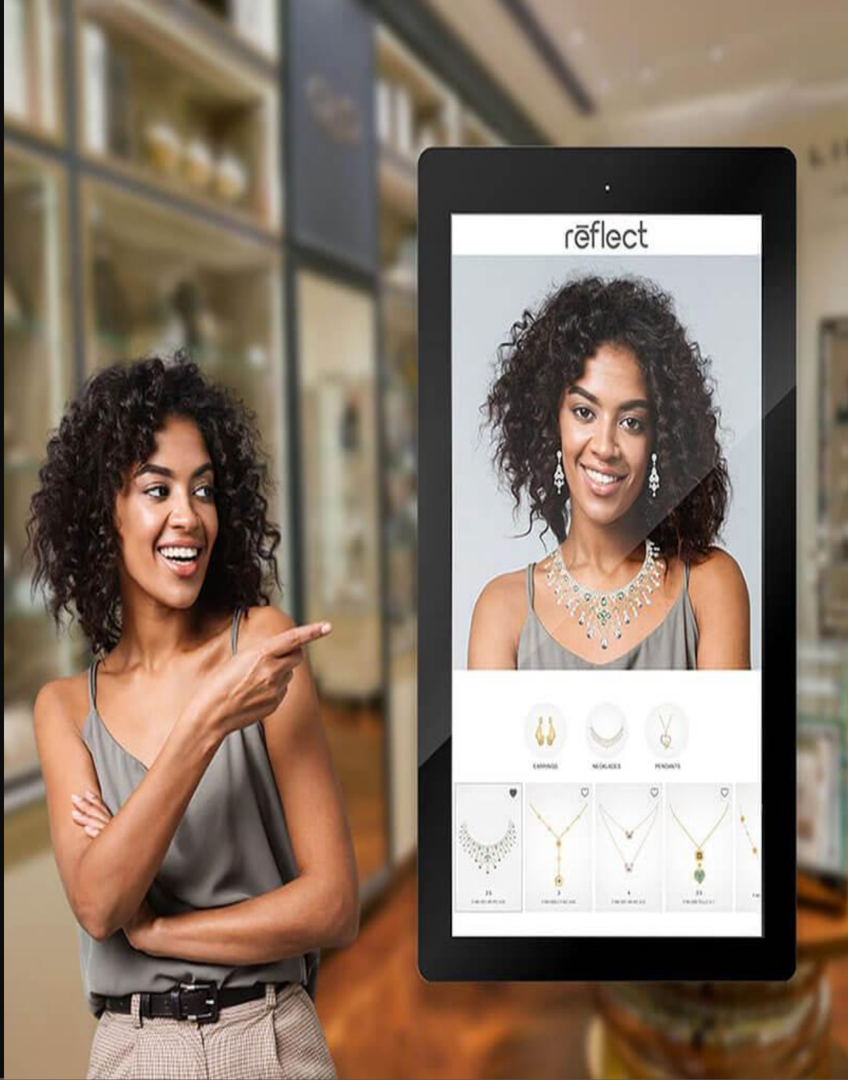

AI/ML Model Scheme

MODEL

- Based on research paper "VITON: An Image-based Virtual Try-on Network."
- Goal: Transfer clothing (c) naturally onto a person in a reference image (I).

VITON Framework: Coarse-to-Fine Approach

1. Clothing-Agnostic Person Representation
   - Contains pose, body parts, face, and hair features.
   - Used as a prior to guide synthesis.

2. Multi-Task Encoder-Decoder Generator
   - Generates a coarse clothed person image and clothing mask.
   - Uses clothing-agnostic person representation and a target clothing image.

3. Thin Plate Spline Transformation
   - Warps clothing (c) to match the person's pose and body shape.

4. Refinement Network
   - Enhances the coarse result with realistic details and deformations.
   - Based on a warped clothing image and a learned composition mask.

Testing and Training

- Utilizes reference image (I) and product image of clothing (c) at test time.
- Network trained with L1 loss for image and mask similarity.
- Inception Score (IS) used for quantitative synthesis quality evaluation.

Rigidity in Pose

- The model may be rigid with unseen poses.
- Flexibility in handling different person poses is a potential challenge.

Model Training and Testing

Testing and Training

- Utilizes reference image (I) and product image of clothing (c) at test time.
- Network trained with L1 loss for image and mask similarity.
- Inception Score (IS) used for quantitative synthesis quality evaluation.

Model used

- Pretrained model was used

Screenshots of user-interfaces and outputs

Original photo

+

Dress

=

Final result